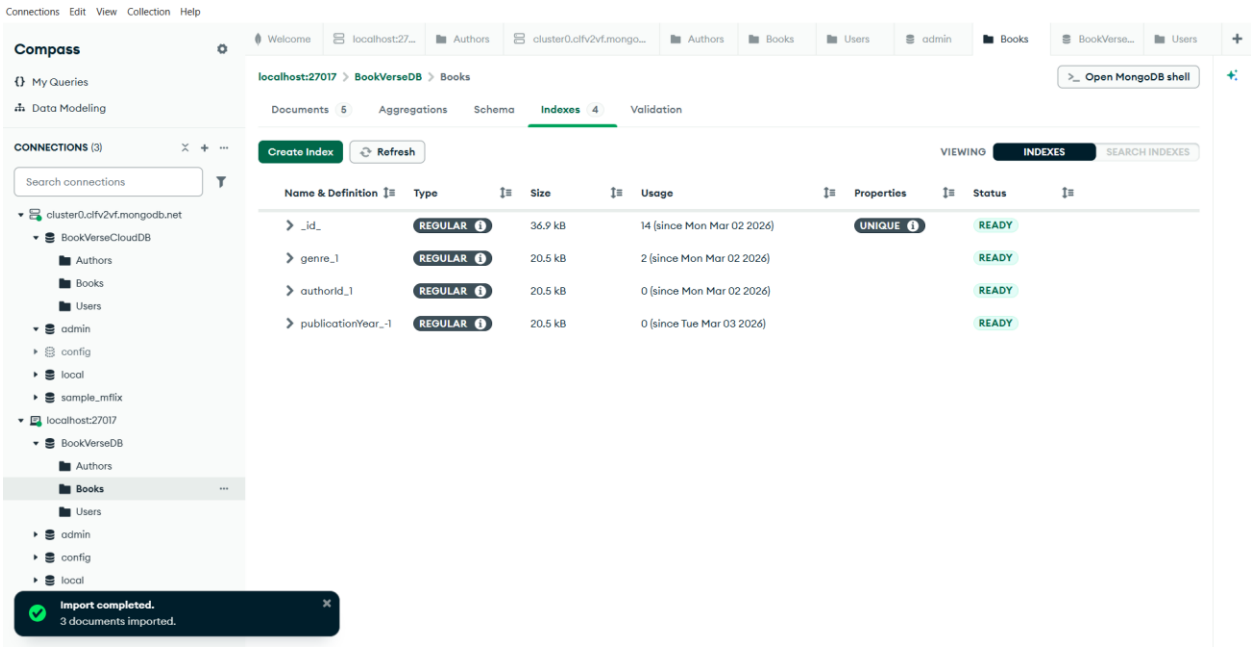
Indexing, Aggregation and MongoDB Atlas Assignment

Your Name :Chandan Kumar

Indexing & Query Optimization

Indexes were created on frequently queried fields: genre, authorId, and ratings.score. Query performance was analyzed using Explain Plan. IXSCAN was observed after index creation and COLLSCAN after dropping an index, demonstrating performance impact.



Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. Learn more

Interpret

Visual Tree | Raw Output

FETCH
Returned 3 Execution Time 0 ms

IXSCAN
Returned 3 Execution Time 0 ms
Index Name: **genre_1**
Multi Key Index: **no**

Query Performance Summary
- **3** documents returned
- **3** documents examined
- **0 ms** execution time
- **Is not** sorted in memory
- **3** index keys examined

Query used the following index:
genre

Close

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. Learn more

Interpret

Visual Tree | Raw Output

FETCH
Returned 2 Execution Time 0 ms

IXSCAN
Returned 2 Execution Time 0 ms
Index Name: **ratings.score_1**
Multi Key Index: **yes**

Query Performance Summary
- **2** documents returned
- **2** documents examined
- **0 ms** execution time
- **Is not** sorted in memory
- **2** index keys examined

Query used the following index:
ratings.score

Close

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. Learn more

Interpret

Visual Tree | Raw Output

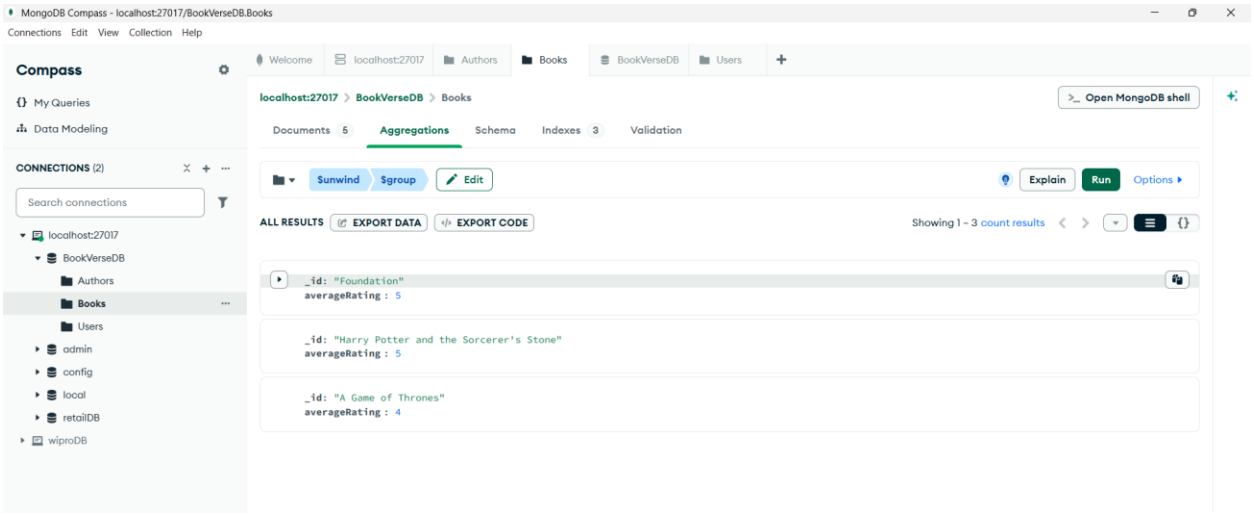
COLLSCAN
Returned 2 Execution Time 0 ms
Documents Examined: **5**

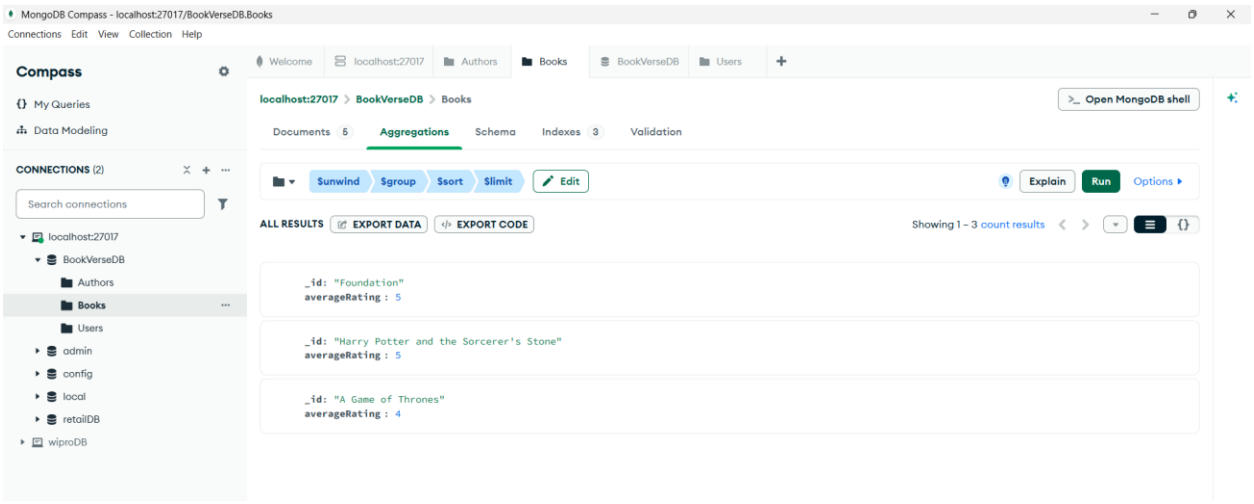
Query Performance Summary
- **2** documents returned
- **5** documents examined
- **0 ms** execution time
- **Is not** sorted in memory
- **0** index keys examined
- No index available for this query.

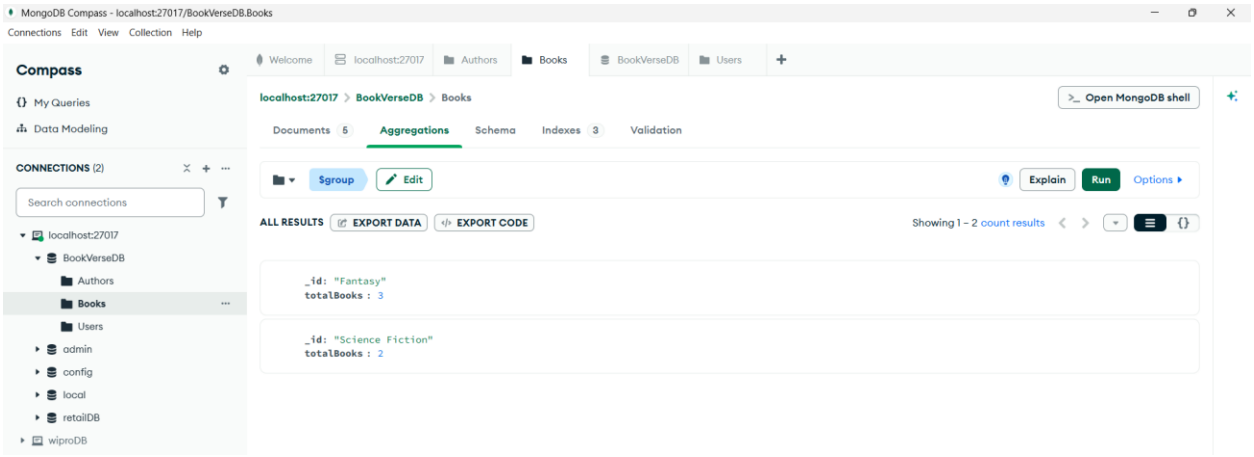
Close

Aggregation Framework

Aggregation pipelines were implemented using $unwind, $group, $avg, $sort, and $limit to generate analytical reports including average ratings, top 3 books, books per genre, and reward points per author.







MongoDB Compass - localhost:27017/BookVerseDB.Books

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

localhost:27017

BookVerseDB

Authors

Books

Users

admin

config

local

retailDB

wiproDB

Welcome | localhost:27017 | Authors | Books | BookVerseDB | Users

localhost:27017 > BookVerseDB > Books

Open MongoDB shell

Documents 5 | Aggregations | Schema | Indexes 3 | Validation

$group | Edit | Explain | Run | Options

ALL RESULTS | EXPORT DATA | EXPORT CODE

Showing 1 – 3 count results

```
_id: ObjectId('69a58a75aa94058da2e9bc4c')
bookCount : 2

_id: ObjectId('69a58a50aa94058da2e9bc48')
bookCount : 2

_id: ObjectId('69a58a64aa94058da2e9bc4a')
bookCount : 1
```

MongoDB Compass - localhost:27017/BookVerseDB.Books

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

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BookVerseDB

Authors

Books

Users

admin

config

local

retailDB

wiproDB

Welcome | localhost:27017 | Authors | Books | BookVerseDB | Users

localhost:27017 > BookVerseDB > Books

Open MongoDB shell

Documents 5 | Aggregations | Schema | Indexes 3 | Validation

$group | $match | Edit | Explain | Run | Options

ALL RESULTS | EXPORT DATA | EXPORT CODE

Showing 1 – 2 count results

```
_id: ObjectId('69a58a75aa94058da2e9bc4c')
bookCount : 2

_id: ObjectId('69a58a50aa94058da2e9bc48')
bookCount : 2
```

MongoDB Compass - localhost:27017/BookVerseDB.Books

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

localhost:27017

BookVerseDB

Authors

Books

Users

admin

config

local

retailDB

wiproDB

Welcome | localhost:27017 | Authors | Books | BookVerseDB | Users

localhost:27017 > BookVerseDB > Books

Open MongoDB shell

Documents 5 | Aggregations | Schema | Indexes 3 | Validation

$unwind | $group | Edit | Explain | Run | Options

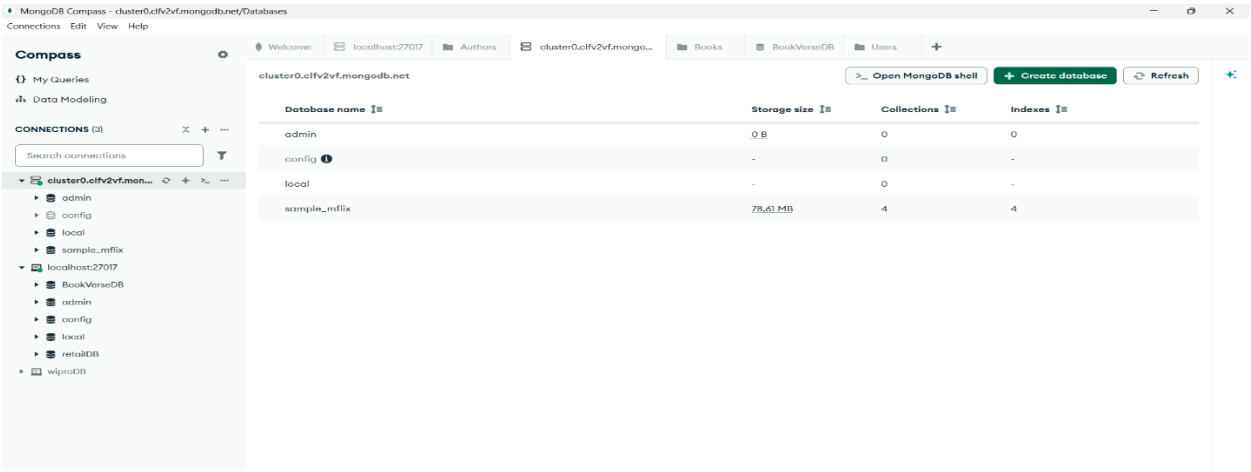
ALL RESULTS | EXPORT DATA | EXPORT CODE

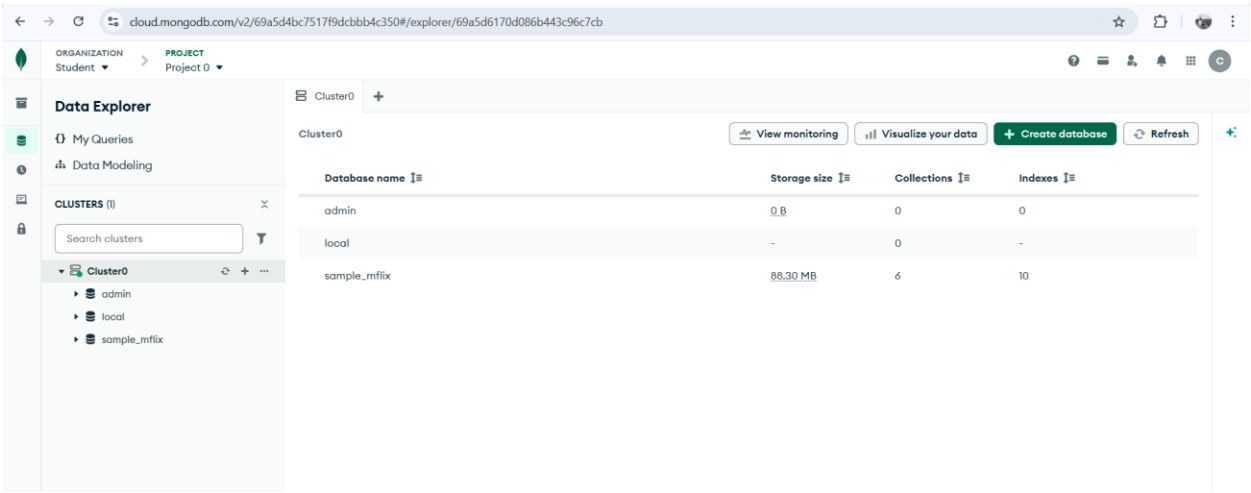
Showing 1 – 3 count results

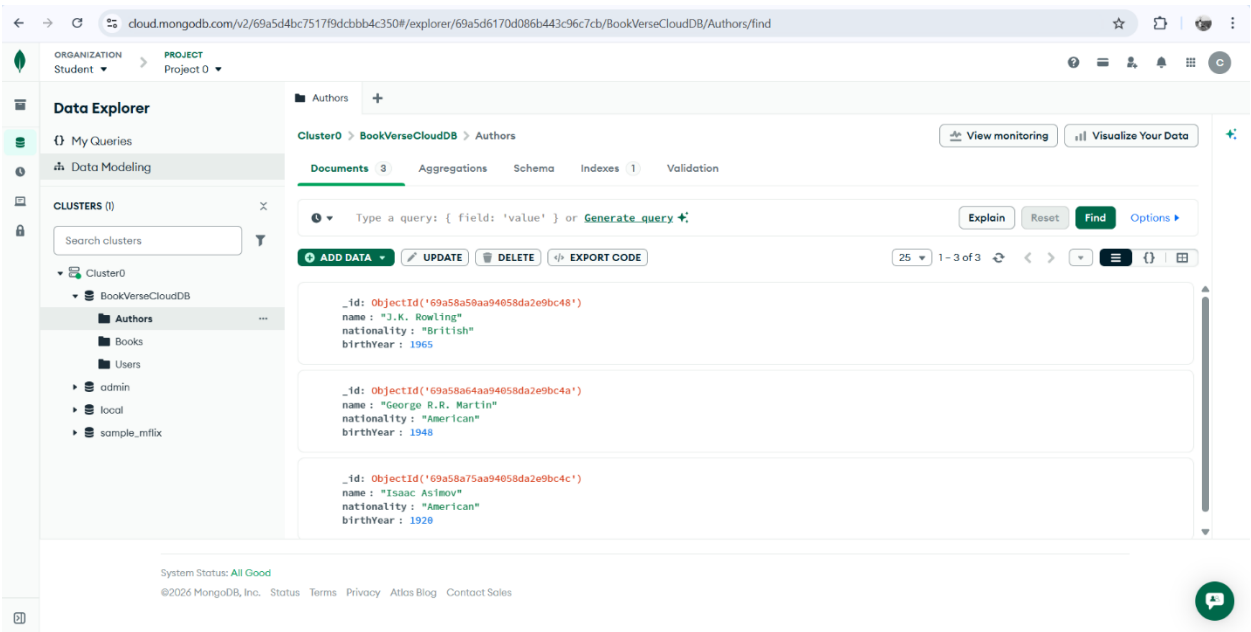
```
_id: ObjectId('69a58a64aa94058da2e9bc4a')
totalRewardPoints : 4

_id: ObjectId('69a58a50aa94058da2e9bc48')
totalRewardPoints : 5

_id: ObjectId('69a58a75aa94058da2e9bc4c')
totalRewardPoints : 5
```

MongoDB Atlas Deployment

A free cluster was created in MongoDB Atlas. BookVerseCloudDB was created and local collections were imported successfully. The database was connected using MongoDB Compass.


MongoDB Compass - cluster0.clfv2vf.mongodb.net/Databases
Connections Edit View Help
Compass
My Queries
Data Modeling
CONNECTIONS (3)
Search connections
cluster0.clfv2vf.mon...
admin
config
local
sample_mflix
localhost:27017
BookVerseDB
admin
config
local
retailDB
wiproDB
Welcome
localhost:27017
Authors
cluster0.clfv2vf.mongo...
Books
BookVerseDB
Users
cluster0.clfv2vf.mongodb.net
Open MongoDB shell
Create database
Refresh
Database name
Storage size
Collections
Indexes
admin 0 B 0 0
config - 0 -
local - 0 -
sample_mflix 78.61 MB 4 4


cloud.mongodb.com/v2/69a5d4bc7517f9dcbbb4c350#/explorer/69a5d6170d086b443c96c7cb
ORGANIZATION
Student
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
admin
local
sample_mflix
Cluster0
View monitoring
Visualize your data
Create database
Refresh
Database name
Storage size
Collections
Indexes
admin 0 B 0 0
local - 0 -
sample_mflix 88.30 MB 6 10


cloud.mongodb.com/v2/69a5d4bc7517f9dcbbb4c350#/explorer/69a5d6170d086b443c96c7cb/BookVerseCloudDB/Authors/find
ORGANIZATION
Student
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
BookVerseCloudDB
Authors
Books
Users
admin
local
sample_mflix
Authors
Cluster0 > BookVerseCloudDB > Authors
View monitoring
Visualize Your Data
Documents 3
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
UPDATE
DELETE
EXPORT CODE
25
1 - 3 of 3
_id: ObjectId('69a58a50aa94058da2e9bc48')
name : "J.K. Rowling"
nationality : "British"
birthYear : 1965
_id: ObjectId('69a58a64aa94058da2e9bc4a')
name : "George R.R. Martin"
nationality : "American"
birthYear : 1948
_id: ObjectId('69a58a75aa94058da2e9bc4c')
name : "Isaac Asimov"
nationality : "American"
birthYear : 1920
System Status: All Good
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Index_Aggregation.txt

Index Creation:

```
db.Books.createIndex({ genre: 1 })

db.Books.createIndex({ authorId: 1 })

db.Books.createIndex({ "ratings.score": 1 })
```

Dropped Index:

```
db.Books.dropIndex("ratings.score_1")
```

Aggregation Queries:

1. Average Rating:

```
[
 { $unwind: "$ratings" },
 { $group: { _id: "$title", averageRating: { $avg: "$ratings.score" } } }
]
```

2. Top 3 Books:

```
[
 { $unwind: "$ratings" },
 { $group: { _id: "$title", averageRating: { $avg: "$ratings.score" } } },
 { $sort: { averageRating: -1 } },
 { $limit: 3 }
]
```

3. Books per Genre:

```
[
 { $group: { _id: "$genre", totalBooks: { $sum: 1 } } }
]
```